

Ex: The scores of two batsman A and B in ten innings during a certain season are as follows:

A	32	28	47	63	71	39	10	60	96	14
B	19	31	48	53	67	90	10	62	40	80

Find which of the two batsman, A and B, is more consistent in scoring.

Sol:

Score of A (x_i)	$u_i = x_i - x_0$ $= x_i - 46$	u_i^2	Score of B (y_i)	$v_i = y_i - y_0$ $= y_i - 50$	v_i^2
32	-14	196	19	-31	961
28	-18	324	31	-19	361
47	1	1	48	-2	4
63	17	289	53	3	9
71	25	625	67	17	289
39	-7	49	90	40	1600
10	-36	1296	10	-40	1600
60	14	196	62	12	144
96	50	2500	40	-10	100
14	-32	1024	80	30	900
Total	0	6500	0	0	5988

$$\bar{u} = \frac{\sum u_i}{n} = 0, \quad \bar{x} = x_0 + \bar{u} = 46$$

$$S.D. = S_x = S_u = \sqrt{\frac{\sum u_i^2}{n} - \bar{u}^2} = \sqrt{\frac{6500}{10} - 0} = 25.5$$

$$\text{Coefficient of variation} = \frac{S_x}{\bar{x}} = \frac{25.5}{46} = 0.5543$$

$$\bar{v} = \frac{\sum v_i}{n} = 0, \quad \bar{y} = y_0 + \bar{v} = 50$$

$$S.D. = S_y = S_v = \sqrt{\frac{5988}{10} - 0} = \sqrt{598.8} = 24.47$$

$$\text{Coefficient of variation} = \frac{S_y}{\bar{y}} = \frac{24.47}{50} = 0.4898$$

Since coefficient of variation of the scores of B is less than that of A, batsman-B is more consistent in scoring.

Ex: Let the mean and standard deviation of a sample of size n_1 with values x_i ($i=1, 2, \dots, n_1$) be denoted by $\bar{x}_1$ and s_1 . Similarly, for another sample of size n_2 with values x_2 ($i=1, 2, \dots, n_2$) by $\bar{x}_2$ and s_2 . Then the variance s^2 of the composite sample is given by

$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2} + \frac{n_1 n_2 (\bar{x}_1 - \bar{x}_2)^2}{(n_1 + n_2)^2}$$

Sol: Let $\bar{x}$ be the mean of the composite set. Now $\bar{x}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} x_{1i}$

$$\text{or, } \sum_{i=1}^{n_1} x_{1i} = n_1 \bar{x}_1$$

$$\text{Also } \bar{x}_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} x_{2i} \quad \text{or, } \sum_{i=1}^{n_2} x_{2i} = n_2 \bar{x}_2$$

$$\text{So } \bar{x} = \frac{\sum_{i=1}^{n_1} x_{1i} + \sum_{i=1}^{n_2} x_{2i}}{n_1 + n_2} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

Also, we have

$$\begin{aligned} \frac{1}{n_1} \sum_{i=1}^{n_1} (x_{1i} - \bar{x})^2 &= \frac{1}{n_1} \sum_{i=1}^{n_1} (x_{1i} - \bar{x}_1 + \bar{x}_1 - \bar{x})^2 \\ &= \frac{1}{n_1} \sum_{i=1}^{n_1} (x_{1i} - \bar{x}_1)^2 + 2(\bar{x}_1 - \bar{x}) \frac{1}{n_1} \sum_{i=1}^{n_1} (x_{1i} - \bar{x}_1) + (\bar{x}_1 - \bar{x})^2 \\ &= S_1^2 + (\bar{x}_1 - \bar{x})^2 \end{aligned}$$

$$\text{since } \frac{1}{n_1} \sum_{i=1}^{n_1} (x_{1i} - \bar{x}_1) = 0$$

$$\text{or, } \sum_{i=1}^{n_1} (x_{1i} - \bar{x})^2 = n_1 S_1^2 + n_1 (\bar{x}_1 - \bar{x})^2$$

$$\text{Similarly, } \sum_{i=1}^{n_2} (x_{2i} - \bar{x})^2 = n_2 S_2^2 + n_2 (\bar{x}_2 - \bar{x})^2$$

But

$$\begin{aligned} S^2 &= \frac{1}{n_1 + n_2} \left[\sum_{i=1}^{n_1} (x_{1i} - \bar{x})^2 + \sum_{i=1}^{n_2} (x_{2i} - \bar{x})^2 \right] \\ &= \frac{1}{n_1 + n_2} [n_1 S_1^2 + n_2 S_2^2 + n_1 (\bar{x}_1 - \bar{x})^2 + n_2 (\bar{x}_2 - \bar{x})^2] \\ &= \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2} + \frac{n_1 (\bar{x}_1 - \bar{x})^2 + n_2 (\bar{x}_2 - \bar{x})^2}{n_1 + n_2} \\ &= \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2} + \frac{1}{n_1 + n_2} \left\{ n_1 \left[\bar{x}_1 - \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \right]^2 + n_2 \left[\bar{x}_2 - \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \right]^2 \right\} \\ &= \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2} + \frac{n_1 n_2 (\bar{x}_1 - \bar{x}_2)^2}{(n_1 + n_2)^2} \end{aligned}$$

$$\text{Thus } \boxed{\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}} \text{ and } \boxed{S^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2} + \frac{n_1 n_2 (\bar{x}_1 - \bar{x}_2)^2}{(n_1 + n_2)^2}}$$

Ex: Consider the data in Table 1. These data are the strengths in pounds per square inch (psi) of 100 glass, nonreturnable 1-liter soft drink bottles. These observations were obtained by testing each bottle until failure occurred

Table 1: Frequency Distribution for the Bursting Strength Data.

Class interval (psi)	Frequency	Relative Frequency	Cumulative Relative Frequency
170 ≤ x < 190	2	0.02	0.02
190 ≤ x < 210	4	0.04	0.06
210 ≤ x < 230	7	0.07	0.13
230 ≤ x < 250	13	0.13	0.26
250 ≤ x < 270	32	0.32	0.58
270 ≤ x < 290	24	0.24	0.82
290 ≤ x < 310	11	0.11	0.93
310 ≤ x < 330	4	0.04	0.97
330 ≤ x < 350	3	0.03	1.00
	100	1.00	

Compute the mean and variance of bursting strength for the data in the above frequency distribution table.

Sol: Note that there are $c=9$ class intervals, and that $x_1=180, f_1=2, x_2=200, f_2=4, x_3=220, f_3=7, x_4=240, f_4=13, x_5=260, f_5=32, x_6=280, f_6=24, x_7=300, f_7=11, x_8=320, f_8=4, x_9=340$ and $f_9=3$. Thus

$$\bar{x} \approx \frac{\sum_{j=1}^9 f_j x_j}{n} = \frac{26,460}{100} = 264.60 \text{ psi}$$

and

$$s^2 \approx \frac{\sum_{j=1}^9 f_j x_j^2}{n} - \frac{1}{n} \left(\sum_{j=1}^9 f_j x_j \right)^2 = \frac{7,091,900}{100} - \frac{(26,460)^2}{100} = 914.99 (\text{psi})^2$$

Ex: The diameter of eight automotive piston rings is shown below. Calculate the sample mean, sample variance and sample standard deviation.

74.001 mm	73.998 mm
74.005	74.000
74.003	74.006
74.001	74.002

Ex: Consider the following frequency distribution. Calculate the sample mean, variance, and standard deviation. Also find the median and mode.

x_i	115	116	117	118	119	120	121	122	123	124
f_i	4	6	9	13	15	19	20	18	15	10

Ex: The mean of two samples of sizes 50 and 100, respectively, are 59.1 and 50.3 and the standard deviations are 8 and 7. Find the standard deviation of the sample of size 150 obtained by combining the two samples.

Ex: The mean of two samples of sizes 50 and 40, respectively, are 59.5 and 54.0 and the standard deviations are 8.38 and 8.23. Find the standard deviation of the sample of size 90 obtained by combining the two samples.

$$\begin{array}{l|l} \text{S:} & \\ \hline n_1 = 50 & n_2 = 40 \\ \bar{x}_1 = 59.5 & \bar{x}_2 = 54.0 \\ s_1 = 8.38 & s_2 = 8.23 \end{array}$$

The mean and s.d. of the combined group are given by

$$\begin{aligned} \bar{x} &= \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \quad \text{and} \quad s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2} + \frac{n_1 n_2 (\bar{x}_1 - \bar{x}_2)^2}{(n_1 + n_2)^2} \\ &= \frac{50 \times 59.5 + 40 \times 54}{50 + 40} \quad \left| \quad \begin{aligned} &= \frac{50 \times (8.38)^2 + 40 \times (8.23)^2}{90} + \frac{50 \times 40 (59.5 - 54)^2}{(90)^2} \\ &= 57.06 \end{aligned} \right. \\ &= 57.06 \quad \left| \quad \begin{aligned} &= 76.59 \\ &\therefore s = \sqrt{76.59} = 8.75 \end{aligned} \right. \end{aligned}$$

Ex: The mean and variance of a group of 100 observations are 6.5 and 3.0 respectively. 55 of these observations have mean 6.6 and standard deviation 1.5. Find the mean and the s.d. of the remaining 45 observations.

Sol: We assume that the 100 observations have been split up into two-groups - Gr. I containing 55 observations with mean $(\bar{x}_1) = 6.6$ and standard deviation $(s_1) = 1.5$, and Gr. II containing 45 observations. The composite group of 100 observations has mean $(\bar{x}) =$

$\bar{x} = 6.5$ and variance (S^2) = 3.0. It is required to find the mean ($\bar{x}_2$) and s.d. (S_2) of Group

Formulae for mean and s.d. of combined group are

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \quad \text{and} \quad S^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2} + \frac{n_1 n_2 (\bar{x}_1 - \bar{x}_2)^2}{(n_1 + n_2)^2} \quad \text{--- (1)}$$

Substituting the values in (1),

$$6.5 = \frac{55 \times 6.6 + 45 \bar{x}_2}{100}$$

$$\text{or, } \bar{x}_2 = 287/45 = \boxed{6.38}$$

$\therefore$ From (1),

$$3.0 = \frac{55 \times (1.5)^2 + 45 S_2^2}{55 + 45} + \frac{55 \times 45 (6.6 - 6.38)^2}{(100)^2}$$

$$\text{or, } 3.0 \times 100 = 55 \times (1.5)^2 + 45 S_2^2 + 1.198$$

$$\text{or, } S_2 = \sqrt{3.89} = \boxed{1.97}$$

Ex: The first four moments of a distribution about the value 4 of the variable are -1.5, 17, -30 and 108. Find the moments about mean, β_1 and β_2 . Find also the moments about (i) the origin, and (ii) the point $x=2$.

Soln: Given $a = 4$ and $m'_1 = -1.5$, $m'_2 = 17$, $m'_3 = -30$ and $m'_4 = 108$.

Moments about mean: $m_1 = 0$

$$m_2 = m'_2 - m'_1{}^2 = 17 - (-1.5)^2 = 17 - 2.25 = 14.75$$

$$\begin{aligned} m_3 &= m'_3 - 3m'_2 m'_1 + 2m'_1{}^3 \\ &= -30 - 3 \times 17 \times (-1.5) + 2(-1.5)^3 \\ &= 39.75 \end{aligned}$$

$$\begin{aligned} m_4 &= m'_4 - 4m'_3 m'_1 + 6m'_2 m'_1{}^2 - 3(m'_1{}^4) \\ &= 108 - 4(-30)(-1.5) + 6(17)(-1.5)^2 - 3(-1.5)^4 \\ &= 142.3125 \end{aligned}$$

$$\text{Hence, } \beta_1 = \frac{m_3}{m_2^{3/2}} = \frac{39.75}{(14.75)^{3/2}} = 0.7016$$

$$\beta_2 = \frac{m_4}{m_2^2} = \frac{142.3125}{(14.75)^2} = 0.6541$$

$$\text{Also, } \bar{x} = a + m'_1 = 4 + (-1.5) = 2.5$$

Moments about origin.

We have $\bar{x} = 2.5$, $m_2 = 14.75$, $m_3 = 39.75$ and $m_4 = 142.31$ (approx.)

$$m'_2 = m_2 + m_1^2 = 14.75 + (2.5)^2 = 21$$

$$m'_3 = m_3 + 3m_2 m_1 + m_1^3 = 39.75 + 3(14.75)(2.5) + (2.5)^3 = 166$$

$$m'_4 = m_4 + 4m_3 m_1 + 6m_2 m_1^2 + m_1^4 = 1132$$

Moments about the point $x=2$: H.W.